

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

04

10 questions · ~15 min

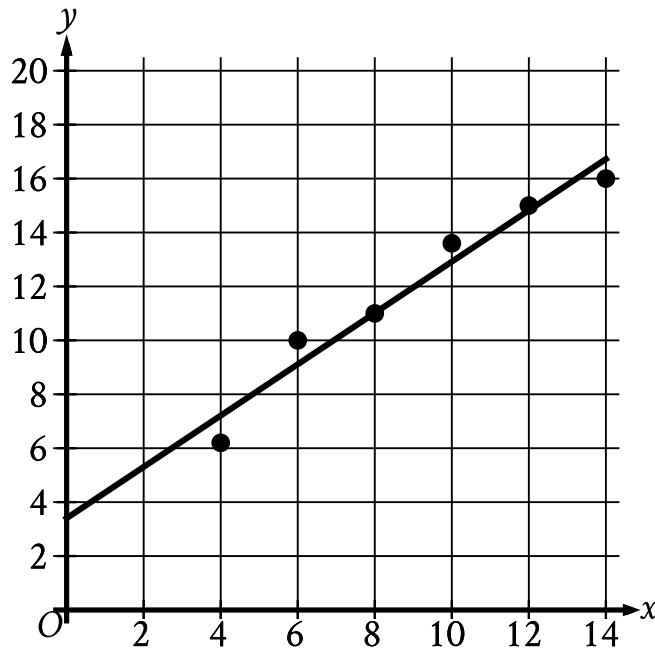
Q1

PROBLEM-SOLVING AND DATA ANALYSIS

For a school fund-raiser, 10 students sold a total of 90 boxes of cookies. Which of the following can be calculated from this information?

- A. The average number of boxes sold per student
- B. The median number of boxes sold per student
- C. The greatest number of boxes sold by one student
- D. The least number of boxes sold by one student

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



- The scatterplot has 6 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
 - The line of best fit slants up from left to right.
 - The line of best fit passes through the following approximate coordinates:
 - (0 comma 3.4)
 - (8 comma 11)
 - (12 comma 14.8)

Which of the following equations best represents the line of best fit shown?

A. $y = x + 3.4$

B. $y = x - 3.4$

C. $y = -x + 3.4$

D. $y = -x - 3.4$

Q3

PROBLEM-SOLVING AND DATA ANALYSIS

The weight of an object on Venus is approximately $\frac{9}{10}$ of its weight on Earth. The weight of an object on Jupiter is approximately $\frac{23}{10}$ of its weight on Earth. If an object weighs 100 pounds on Earth, approximately how many more pounds does it weigh on Jupiter than it weighs on Venus?

A. 90

B. 111

C. 140

D. 230

Q4

PROBLEM-SOLVING AND DATA ANALYSIS

Scott selected 20 employees at random from all 400 employees at a company. He found that 16 of the employees in this sample are enrolled in exactly three professional development courses this year. Based on Scott's findings, which of the following is the best estimate of the number of employees at the company who are enrolled in exactly three professional development courses this year?

- A. 4
- B. 320
- C. 380
- D. 384

Q5

PROBLEM-SOLVING AND DATA ANALYSIS

The cost of a certain shirt is \$20 before a 5% sales tax is added. What is the total cost, including sales tax, to purchase the shirt?

- A. \$20.05
- B. \$20.50
- C. \$21.00
- D. \$25.00

There are n nonfiction books and 12 fiction books on a bookshelf. If one of these books is selected at random, what is the probability of selecting a nonfiction book, in terms of n ?

A. $\frac{n}{12}$

B. $\frac{n}{n+12}$

C. $\frac{12}{n}$

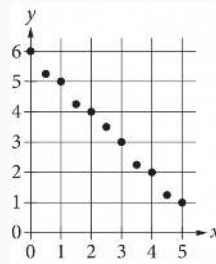
D. $\frac{12}{n+12}$

Number of High School Students Who
Completed Summer Internships

High school	Year				
	2008	2009	2010	2011	2012
Foothill	87	80	75	76	70
Valley	44	54	65	76	82
Total	131	134	140	152	152

The table above shows the number of students from two different high schools who completed summer internships in each of five years. No student attended both schools. Which of the following statements are true about the number of students who completed summer internships for the 5 years shown?

1. The mean number from Foothill High School is greater than the mean number from Valley High School.
 2. The median number from Foothill High School is greater than the median number from Valley High School.
- A. I only
- B. II only
- C. I and II
- D. Neither I nor II



Which of the following could be an equation for a line of best fit for the data in the scatterplot?

- A. $y = -x + 6$
- B. $y = -x - 6$
- C. $y = 6x + 1$
- D. $y = 6x - 1$

A cherry pitting machine pits 12 pounds of cherries in 3 minutes. At this rate, how many minutes does it take the machine to pit 96 pounds of cherries?

- A. 8
- B. 15
- C. 24
- D. 36

A company designs and makes handbags. To estimate the mean weight of the handbags made by the company on a particular day, a sample of the handbags made by the company on that day was selected at random. Based on the sample, it is estimated that the mean weight of all handbags made by the company on that day is 27.8 ounces oz, with an associated margin of error of 0.02 oz. Based on this estimate and associated margin of error, which of the following is the most plausible conclusion?

- A. The mean weight of all handbags made by the company on that day is between 27.78 oz and 27.82 oz.
- B. The actual weights of all handbags made by the company on that day are between 27.78 oz and 27.82 oz.
- C. The actual weights of all handbags from the sample are between 27.78 oz and 27.82 oz.
- D. The mean weight of all handbags made by the company on that day is 27.8 oz.